

# Hua Yang

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## Education

NC State University, Raleigh, NC Ph.D., Computer Science GPA: **3.8/4.0** Aug 2024 – Expected Dec 2027  
South China University of Technology, China M.S., Computer Science GPA: **4.0/4.0** Sep 2021 – Jun 2024  
Central South University, China B.S., Electronic Engineering GPA: **4.0/4.0** Sep 2017 – Jun 2021

## Skills

**ML/LLM Systems:** PyTorch, CUDA, DeepSpeed, Ray, vLLM, Hugging Face

**Agent Systems:** Agent Evaluation, ReAct, MCP, RAG, LangGraph, LangSmith, Agentic Workflows

**Infrastructure:** Kubernetes, Docker, AWS, CI/CD, REST APIs, SQL

**Languages:** Python, Go, Java, JavaScript

## Work Experience

**Software Engineer Intern, AI Infrastructure** ByteDance, San Jose, CA *May 2026 – Aug 2026*

- Built an offline evaluation framework for multi-turn on-call agents by converting production traces into causal graphs, achieving **35% higher conversion success** than a Claude-based baseline.
- Extended agent infrastructure with a **real-device rendering and execution harness**, enabling end-to-end, verifiable evaluation of agent-generated remediation workflows in realistic application environments.
- Developed an LLM-driven data synthesis pipeline that emulates software engineer interactions over sanitized production data, generating **1,000 evaluation-ready test cases**.

**Machine Learning Intern** Pazhou Lab, Guangzhou, China *May 2022 – May 2023*

- Built a sleep disorder detection system, ranking **5th/171 nationally** and securing a \$10,000 award.
- Improved model accuracy from **55% to 59%** across 7 datasets while achieving a **110× runtime speedup**.

## Projects

**Causal Analysis of LLM Memorization & Membership Inference** *Sep 2024 – May 2025*

- **Tech:** SFT, Causal Inference, DeepSpeed, PyTorch, Hugging Face
- Fine-tuned code LLMs on semantically transformed datasets with **<1.5% accuracy loss** while reducing membership inference risk by up to **10.19%**.
- Built structural causal models to quantify the causal effects of code complexity, length, and transformations on memorization; published at **ICSE 2026**.

**GDPR-Compliant Code LLM Alignment & Agent Design** *Jan 2026 – May 2026*

- **Tech:** Ray, vLLM, DPO, CodeLlama, Multi-Agent Systems
- Built an automated GitHub PR collection and annotation pipeline with **1,173 labeled GDPR preference samples**.
- Designed a **Generator-Challenger-Judge** multi-agent framework with semantic memory and self-evolving pattern extraction, improving detection from **35% to 65%**.
- Implemented distributed **DPO** training with Ray and vLLM to align CodeLlama with GDPR compliance preferences.

**Causal Analysis of Training Dynamics & PII Leakage in Code LLMs** *August 2025 – Jan 2026*

- **Tech:** LoRA, Hugging Face, DoWhy, Causal Inference, Dataset Cartography
- Built a real-world PII dataset of **9,000 code files across 6 PII types** using regex, LLM-based refinement, heuristics, and expert validation.
- Fine-tuned **6 code LLMs (3B–14B)** across Stable Code, CodeLlama, and Qwen2.5-Coder, analyzing training dynamics and black-box PII leakage across model families.
- Built structural causal models with DoWhy to quantify how learning difficulty affects privacy leakage, establishing causal links between **LLM training dynamics and PII exposure**.

## Selected Publications (First Author)

- *How Do Semantically Equivalent Code Transformations Impact Membership Inference on LLMs for Code?* **ICSE 2026** (CORE A\*, top-tier SE conference)
- *A Causal Perspective on the Role of Training Dynamics for Interpreting Privacy Risks in Code Models.* **ACM TOSEM** (TOSEM, flagship SE journal)
- *Improving the Interpretability through Maximizing Mutual Information for EEG Emotion Recognition.* **IEEE TAC**